Course Project for SE

Front-End



Group Name: Power Puff Girls

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Instructor Name:

Dr. Shahid Manzoor

Section: BSDSF24M

Department of Data Science

Punjab University, Lahore

Project Deliverable D2

Requirements Elicitation

Ludo Game Platform (Front End)

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Roll number	Team member
BSDSF24M004	Zainab Tariq
BSDSF24M022	Saneela Aziz
BSDSF24M034	Hifza Zia

1. Scope statement

This document outlines the requirements for the front end of the Ludo Game Platform (LGP), the game platform component of the Ludo project developed by team Power Puff Girls (Zainab Tariq, Saneela Aziz and Hifza Zia) during Sprint 1 of the Software Engineering course in Fall 2026. It presents the user requirements elicited from the client and is organised into the following sections: the scope statement (Section 1), the user classes described as user personas (Section 2), and the requirements written as user stories (Section 3), followed by the changelog.

The client for this project (played by our teacher and TA) wants a digital multiplayer board-race game for 2 to 4 players, inspired by traditional race-based board games such as Ludo. Each player has several pieces and wins by moving all of them from the starting area to the final destination before the other players. The game can be played human vs human or human vs computer, on desktop or in a web browser.

The front end provides everything the players see and use: game setup, the board, pieces and dice, turn and progress information, messages for valid and invalid moves, the winner screen and the restart options. The game rules, move validation and the computer opponent are handled by the game logic and game AI components, which the front end is expected to use but which are outside the scope of this document.

2. User classes

The front end has three types of end users: casual players, competitive players and first-time players. Each is described below as a user persona representative of the people who will use the product.

2.1 Rida - Casual player

Rida, 20, is a second-year university student. She does not consider herself a gamer, but she likes simple board games she can play in short breaks between classes, usually on her laptop or phone. She grew up playing Ludo with her cousins and wants a digital version she can start in a few clicks without reading long instructions. Most of the time she plays against the computer, and sometimes she plays with friends on the same device.

Rida dislikes cluttered screens. She expects a clear board, a big dice button and an obvious sign of whose turn it is, and she wants the game to tell her immediately when a move is not allowed.

2.2 Hamza - Competitive player

Hamza, 24, is a software developer who enjoys strategy games and friendly competition. He plays several times a week in the evenings and likes to see the state of all players at a glance: how many pieces each player has at home, on the track and finished. He often plays against the computer to practise and expects it to make sensible moves quickly.

Hamza expects the rules to be enforced strictly and the dice to be fair. He finds long animations annoying, so he wants to be able to skip or switch them off. He wants to restart or start a new game quickly after each match.

2.3 Maryam - First-time player

Maryam, 12, is a school student who has never played a race board game. She is learning from her older brother and is easily discouraged by error messages. She needs the interface to guide her: which pieces can move, where they will land, and what the rules are.

Maryam prefers short, simple wording, colours and visual hints to long text. She wants to read a short rules summary before the game and during it, and she wants to be told in friendly language why a move is not allowed.

3. User requirements

The requirements below were elicited from the client during Sprint 1. Functional requirements are numbered FR-01 onward and non-functional requirements NFR-01 onward, as asked by our teacher and TA. They are written as user stories in the format "As a <role>, I want <goal>, so that <benefit>" (non-functional requirements use "The game shall ..."). Items marked *(to confirm)* depend on rules the client has not yet specified.

3.1 Game setup and modes

ID	User story
FR-01	As a player, I want to start a new game from the main screen, so that I can begin playing quickly.
FR-02	As a player, I want to choose 2, 3 or 4 players, so that the game fits the people available.
FR-03	As a player, I want a Human vs Human mode, so that I can play against other people.
FR-04	As a player, I want a Human vs Computer mode, so that I can play when no other people are available.
FR-05	As a player, I want to mark individual seats as computer-controlled, so that humans and computer players can play together *(to confirm)*.
FR-06	As a player, I want to enter a name and pick a colour, so that everyone can recognise my pieces.
FR-07	As a first-time player, I want to read a short rules summary before and during the game, so that I can understand how to play.

3.2 Turns and dice

ID	User story
FR-08	As a player, I want the system to decide who plays first, so that the game starts fairly *(to confirm how: highest roll or fixed order)*.

ID	User story
FR-09	As a player, I want turns to pass in a fixed order with the active player clearly shown, so that I always know when to act.
FR-10	As a player, I want to roll the dice with one click or tap, so that my turn is quick.
FR-11	As a player, I want the dice result to be random and clearly displayed, so that the game is fair and I know how far I can move.
FR-12	As a player, I want a special roll (for example a six) to give an extra turn or release a piece from the starting area, so that the game has strategy *(to confirm which roll and which effect)*.
FR-13	As a player, I want my turn to be skipped automatically when I have no valid move, so that the game keeps flowing.

3.3 Movement and valid moves

ID	User story
FR-14	As a player, I want the pieces that can legally move to be highlighted after I roll, so that I do not have to work out the options myself.
FR-15	As a player, I want to select a piece and move it by the number rolled, so that moving is simple.
FR-16	As a player, I want the system to allow only valid moves, so that nobody can break the rules.
FR-17	As a player, I want to see a clear message explaining why a move is invalid, so that I can choose another move.
FR-18	As a player, I want to see where a piece will land before I confirm the move, so that I can decide with confidence.
FR-19	As a player, I want to see pieces move step by step along the track, so that I can follow what happened.
FR-20	As a player, I want the rule for entering the final destination (exact roll needed or not) to be enforced automatically *(to confirm)*.

3.4 Interaction between players

ID	User story
FR-21	As a player, I want to capture an opponent's piece by landing on it, sending it back to its starting area, so that I can compete with my opponents *(to confirm)*.
FR-22	As a player, I want some squares to be safe from capture and clearly marked, so that I can plan my moves *(to confirm)*.
FR-23	As a player, I want to be told when a piece is captured, so that I understand why it moved.
FR-24	As a player, I want the rules for two pieces on the same square (stacking or blocking) to be applied automatically *(to confirm)*.

3.5 Progress tracking and winning

ID	User story
FR-25	As a player, I want the system to track turns and each player's progress automatically, so that no manual counting is needed.
FR-26	As a player, I want a status panel showing, for each player, how many pieces are in the starting area, on the track and at the destination, so that I know who is winning.

ID	User story
FR-27	As a player, I want the system to detect the winner automatically when all of a player's pieces reach the destination, and show an end screen, so that the result is immediately clear.
FR-28	As a player, I want the final ranking of the other players to be shown after the winner is decided *(to confirm)*.
FR-29	As a player, I want to see a short session score during consecutive rematches, so that I know who is leading in a series of games.

3.6 Computer player

ID	User story
FR-30	As a player, I want a computer player to choose and make a valid move on its turn, so that I can play alone.
FR-31	As a player, I want to choose the computer's difficulty level, so that the game matches my skill *(to confirm)*.

3.7 Game control

ID	User story
FR-32	As a player, I want to restart the current game, so that I can start over.
FR-33	As a player, I want to start a new game with new settings after one ends, so that we can keep playing.
FR-34	As a player, I want to be asked for confirmation before leaving or restarting a game, so that I do not lose progress by accident.

3.8 Interface and settings

ID	User story
FR-35	As a player, I want to see the board with starting areas, track and destination, the pieces, the dice, player status and game information on one screen, so that I understand the game at a glance.
FR-36	As a player, I want the last action (for example "Hifza rolled 4 and captured a piece") shown in a message area, so that I do not miss what other players did.
FR-37	As a player, I want to turn sound effects on or off, so that I can play in quiet places.
FR-38	As a player, I want to skip or switch off animations, so that the game runs as fast as I want.
FR-39	As a player, I want to switch between a light and a dark theme, so that I can play comfortably in different lighting conditions.

3.9 Non-functional requirements

ID	Requirement
NFR-01	Performance: the game shall respond to every user action (click, roll, move) within 1 second *(value to confirm)*.
NFR-02	Performance: the computer player shall choose its move within 2 seconds *(value to confirm)*.
NFR-03	Reliability: the game shall not crash during normal gameplay.

ID	Requirement
NFR-04	Reliability: the game shall handle invalid moves and unexpected input without losing the current game state.
NFR-05	Fairness: the dice shall use an unbiased random number generator, so that every outcome is equally likely.
NFR-06	Usability: a new player shall be able to understand the rules and complete a first turn without outside help *(measure to confirm)*.
NFR-07	Usability: text, pieces and board squares shall be readable on the supported screens, and colours shall be distinguishable by shape or label as well as colour.
NFR-08	Usability: rules and messages shall use short, simple wording.
NFR-09	Usability: the game shall be playable with the keyboard as well as the mouse.
NFR-10	Platform: the game shall run on desktop in the latest versions of Chrome, Firefox and Edge without installation *(browsers to confirm)*.
NFR-11	Platform: the layout shall adapt to different window sizes *(to confirm whether mobile is required)*.
NFR-12	Extensibility: game rules and settings (number of pieces, board size, safe squares, special rolls) shall be kept in configuration, separate from the interface code, so that rule changes do not need a redesign.
NFR-13	Extensibility: the front end shall communicate with the game logic and the computer player through a defined interface, so that each can be replaced or extended independently.
NFR-14	Maintainability: the code shall be structured and commented so that all three team members can understand and modify it.

4. Changelog

This is the first version of the document, so there are no changes with respect to a previous deliverable.

Version	Date	Change
v1	2026-10-18	Initial version